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Analysis of Branched Cracks

This paper presents a method for solving a class of two-dimensional elastic branched crack problems. In contrast to other approaches in the literature, the integral equation presented here enables different branched crack problems to be solved in a unified manner. Muskhelishvili's potential formulation is used to derive, by means of a Green's function technique, a singular integral equation in complex form. The problems of the asymmetrically, symmetrically, and doubly symmetrically branched cracks are considered. The ratio of the length of the branched crack to that of the main one may be varied arbitrarily and the limit in which this ratio goes to zero is obtained analytically. Stress-intensity factors at the branched crack tip are computed numerically and the results, where possible, are compared to those in the literature. Disagreements in the literature are discussed and clarified with the aid of the present results.

Introduction

Among the different configurations of branched cracks, the asymmetrically (singly) branched crack, shown in Fig. 1, has received most attention in the literature. There have been many attempts [1-13] to solve this problem for arbitrary values of c/l , the ratio of the half-length c of the main crack to the length l of the branch. Of particular physical interest is the limiting case as $c/l \rightarrow \infty$, where the solution [3, 4, 6, 7] has been used to predict the initial angle of branching of a crack in brittle solids under mixed-mode loading. For this type of loading, the angle that gives the greatest energy release rate at the infinitesimal branched tip has generally been assumed to be the angle at which the main crack will initiate. Other criteria [14, 15] have also been proposed to predict quantitatively this angle of bifurcation. A review in this area was given by Palaniswamy and Knauss [3].

With a few exceptions [4, 9, 10], most of the analyses of these branched crack problems were based on the Muskhelishvili potential formulation and conformal mapping of the branched crack geometry onto the unit circle. The solution of the problem is then reduced to either infinite series expansions or integral equations. Bilby and Cardew [4], in a paper that made use of a solution by Khrapkov [5],

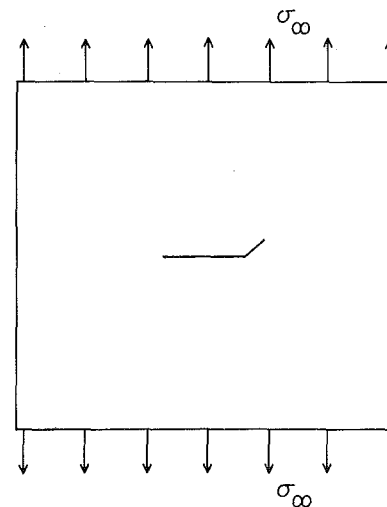


Fig. 1 Asymmetrically branched crack

calculated the greatest energy release rate at the branched crack tip for the limiting case $c/l \rightarrow \infty$. Palaniswamy and Knauss [3], despite the approximate nature of their method of infinite series expansions, obtained results in agreement with those of [4]. Dudukalenko and Romalis [6], Hussain, et al. [7], independently derived analytical expressions for the complex potentials, but their results disagreed with those of the previously mentioned authors. Bui [16] noted the disagreement between [6, 7] but he did not elaborate on it any further. Wu [13] considered the problem of a Z-crack and obtained approximate results for the limit $c/l \rightarrow \infty$. In this limit the solution is the same

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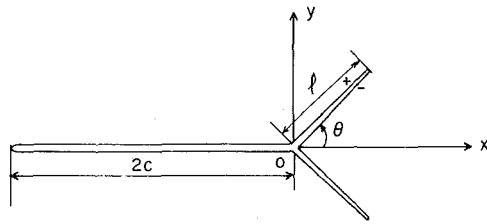


Fig. 2 Symmetrically branched crack

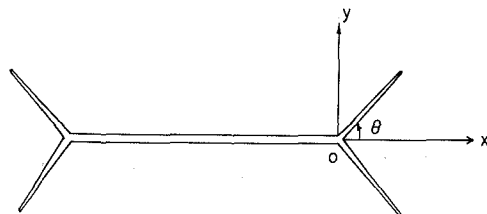


Fig. 3 Doubly symmetrical branched crack

as that of the singly branched crack. For finite c/l , Chatterjee [8], Gupta [9], Kitagawa, et al. [11, 12], have studied this single branch crack problem independently and their results agree with each other. We note also that almost all of the studies for the limit $c/l \rightarrow \infty$ cannot be applied to the analysis of a finite c/l and vice versa.

Theocaris and Ioakimidis [17], using the theory developed by Datsyshin and Savruk [18], reported a solution of the symmetrically branched crack for finite c/l . Bilby, Cardew, and Howard [10], Kitagawa, Yuuki, and Ohira [11, 12], Cherepanov and Kuliev [13] have all analyzed this problem. Bilby, et al. [10], concluded that the solution in [21] is incorrect, contending that the Wiener-Hopf equations derived for the problem ought to be coupled. However, the results in [10], in turn, do not agree with [11, 12]. In all of the foregoing analyses, the limiting case of $c/l \rightarrow \infty$ either remains inaccessible or has not been studied.

The doubly symmetrical branched crack (Fig. 3) was studied by Kitagawa, et al. [11, 12], for finite c/l .

This paper attempts to settle conclusively the disagreements in the literature by solving the different branched-crack problems in a unified way with the aid of a general integral equation method based on Green's functions and Muskhelishvili's complex potentials. The method generalizes the work of Greif and Sanders [19] and Rice [20]. In contrast to most of the previous work, the case of finite c/l and the limit $c/l \rightarrow \infty$ can be handled within the same formulation.

Formulation

Consider the problem of elastic branched cracks in a two-dimensional infinite medium (Figs. 1–3). Uniform tensile stress is prescribed at infinity. The problem is formulated in terms of the complex potentials $\tilde{\phi}, \tilde{\psi}$ of Muskhelishvili [22]. The stresses and the displacements can be expressed in terms of these analytic functions $\tilde{\phi}, \tilde{\psi}$ as

$$\begin{aligned}\tilde{\sigma}_{xx} + \tilde{\sigma}_{yy} &= 2(\tilde{\phi}' + \overline{\tilde{\phi}'}), \quad \tilde{\sigma}_{yy} - \tilde{\sigma}_{xx} + 2i\tilde{\sigma}_{xy} = 2(\tilde{z}\tilde{\phi}'' + \overline{\tilde{\psi}'}), \\ 2G(\tilde{u}_x + i\tilde{u}_y) &= \kappa\tilde{\phi} - \tilde{z}\tilde{\phi}' - \overline{\tilde{\psi}},\end{aligned}\quad (1)$$

where $\tilde{\sigma}_{xx}, \tilde{\sigma}_{yy}$ are the normal stress components in the $\tilde{x}$ and $\tilde{y}$ directions, $\tilde{\sigma}_{xy}$ is the shear stress, $\tilde{u}_x$ and $\tilde{u}_y$ are the Cartesian components of displacement, G is the shear modulus, and $\kappa = 3 - 4\nu$ for plane strain, $\kappa = (3 - \nu)/(1 + \nu)$ for plane stress, ν being Poisson's ratio. The potentials $\tilde{\phi}, \tilde{\psi}$ are analytic functions of the complex variable $\tilde{z} = \tilde{x} + i\tilde{y}$, except along appropriate branch cuts in the $\tilde{z}$ -plane. The prime denotes differentiation with respect to $\tilde{z}$ and a bar over a functional symbol indicates complex conjugation.

We now introduce the dimensionless variables

$$(\tilde{\sigma}_{xx}, \tilde{\sigma}_{yy}, \tilde{\sigma}_{xy}) = \sigma_{\infty}(\sigma_{xx}, \sigma_{yy}, \sigma_{xy}), \quad (\tilde{\phi}, \tilde{\psi}) = \sigma_{\infty}l(\phi, \psi), \quad \tilde{z} = lz, \quad (2)$$

where l is the length of the branch, c is the half-length of the main crack, and σ_{∞} is the magnitude of the prescribed tensile stress at infinity. To insure uniform stress at infinity, the dimensionless potentials ϕ', ψ' have to behave asymptotically as follows:

$$\phi' \sim \frac{1}{4}, \quad \psi' \sim \frac{1}{2} \quad \text{as } |z| \rightarrow \infty. \quad (3)$$

Primes now imply differentiation with respect to the dimensionless variable z .

In the limiting case as $c/l \rightarrow \infty$, the stress distribution at infinity becomes the asymptotic elastic crack-tip field given by

$$\sigma_{ij}^{(\infty)} \sim \frac{K_I^{(\infty)} f_{ij}(\varphi)}{\sqrt{2\pi\rho}} \quad \text{as } \rho = |z| \rightarrow \infty, \quad (4)$$

where $K_I^{(\infty)}$, the prescribed Mode I intensity factor, is regarded as the parameter characterizing the applied load. The functions $f_{ij}(\varphi)$ are known functions of the polar angle φ , and ρ is the distance from the crack tip. The appropriate dimensionless variables are then

$$\begin{aligned}(\tilde{\sigma}_{xx}, \tilde{\sigma}_{yy}, \tilde{\sigma}_{xy}) &= K_I^{(\infty)}/\sqrt{l}(\sigma_{xx}, \sigma_{yy}, \sigma_{xy}), \\ (\tilde{\phi}, \tilde{\psi}) &= K_I^{(\infty)}\sqrt{l}(\phi, \psi), \quad \tilde{z} = lz, \text{ etc.}\end{aligned}\quad (5)$$

The asymptotic forms of the potentials ϕ', ψ' corresponding to (4) are

$$\phi' \sim \frac{1}{2\sqrt{2\pi z}}, \quad \psi' \sim \frac{1}{4\sqrt{2\pi z}} \quad \text{as } |z| \rightarrow \infty. \quad (6)$$

According to (1), the boundary conditions for a stress-free main crack, which lies on the real axis, are

$$\sigma_{yy} + i\sigma_{xy} = \phi' + \overline{\phi'} + x\phi'' + \psi' = 0. \quad (7)$$

Singular Solutions

A crack may be viewed as a continuous distribution of dislocation singularities. Thus it is necessary to study first the singular solution corresponding to a dislocation line. The elastic stress field due to an edge dislocation is well known [20]. In the context of Muskhelishvili's complex-potential formulation, the dislocation singularity appears as a branch point in the complex plane. The potentials ϕ_0, ψ_0 corresponding to a single edge dislocation line at z_0 in an infinite medium are given by

$$\begin{aligned}\phi_0 &= f_1(z, z_0, \alpha) = \alpha \ln(z - z_0), \\ \psi_0 &= g_1(z, z_0, \alpha) = \overline{\alpha} \ln(z - z_0) - \alpha \overline{z}_0/(z - z_0),\end{aligned}\quad (8)$$

where $\alpha = G\mu e^{i\theta}/\pi i(\kappa + 1)$, $\mu = [u_r] + i[u_\theta]$, $[u_r] = u_r^+ - u_r^-$, etc. The quantities $[u_r], [u_\theta]$ are the jumps in the tangential and normal displacements across the dislocation line. The aforementioned expressions for ϕ_0, ψ_0 are derived by imposing zero net traction and moment in a circuit around the dislocation, and by demanding that the jumps in the displacements be given by $[u_r]$ and $[u_\theta]$. Furthermore, the stresses are required to be no more singular than $1/|z - z_0|$ near z_0 .

In the presence of a crack, the solution for an edge dislocation is no longer given by ϕ_0, ψ_0 alone. In addition to the conditions imposed at z_0 , the boundary conditions for the crack have to be satisfied. The new potentials ϕ_1, ψ_1 for an edge dislocation in the presence of a crack may now be written in the form

$$\phi_1 = \phi_0 + \phi_R, \quad \psi_1 = \psi_0 + \psi_R, \quad (9)$$

where ϕ_R, ψ_R are to be determined in such a way that the stresses on the crack due to ϕ_1, ψ_1 are zero. Equivalently, the problem may be reduced to finding ϕ_R, ψ_R with prescribed stresses on the crack equal to the negative of those induced by ϕ_0, ψ_0 . The solution of this problem is known [22] and was summarized by Erdogan [23]. Since, near z_0 , the potentials ϕ_1, ψ_1 must still have the same singular behavior as ϕ_0, ψ_0 , it follows that the unknowns ϕ_R, ψ_R are regular at z_0 . Substituting for ϕ_0, ψ_0 from (8) into (7), we obtain the stresses on the crack induced by the edge dislocation alone at z_0 in the form

$$\sigma_{yy}^{(0)}(x, 0) - i\sigma_{xy}^{(0)}(x, 0) = \frac{\alpha}{x - z_0} + \frac{\alpha}{x - \bar{z}_0} + \frac{\alpha(z_0 - \bar{z}_0)}{(x - \bar{z}_0)^2} = -p(x, z_0) \quad (10)$$

where the superscript (0) is to indicate that the stresses are derived from ϕ_0, ψ_0 by means of (7). From formulas in [23], we have the representation

$$\phi_R'(z) = \frac{1}{2\pi i X(z)} \int_L \frac{X(t)p(t, z_0)dt}{t - z} + \frac{C_1}{X(z)}, \quad (11)$$

where $X(z) \equiv \sqrt{z(z + 2c)}$. The branch of $X(z)$ is taken such that $X(z) \sim z$ as $|z| \rightarrow \infty$. The constant C_1 is as yet undetermined. Through contour integration and results in [22, 23], together with the requirement that the displacement be single-valued, one is finally led to

$$\left. \begin{aligned} \phi_R'(z) &= -[\alpha F(z, z_0) + \alpha F(z, \bar{z}_0) + \bar{\alpha}(z_0 - \bar{z}_0)G(z, \bar{z}_0)], \\ \psi_R'(z) &= \bar{\phi}_R'(z) - \phi_R'(z) - z\phi_R'', \end{aligned} \right\} \quad (12)$$

where

$$F(z, z_0) \equiv \frac{1}{2} \left[1 - \frac{X(z_0)}{X(z)} \right] / (z - z_0),$$

$$G(z, z_0) \equiv \frac{\partial}{\partial z_0} F(z, z_0) = \frac{1}{2} \left[1 - \frac{X(z_0)}{X(z)} - (z - z_0) \frac{X'(z_0)}{X(z)} \right] / (z - z_0)^2, \quad C_1 = -\alpha \quad (13)$$

We shall see that there is no need to find the functions ϕ_R, ψ_R explicitly since the stresses are determined entirely by the derivatives of ϕ_R and ψ_R . In the limit as $c/l \rightarrow \infty$, the Green's function $F(z, z_0)$ in (13) approaches the limit given by

$$F(z, z_0) = \frac{1}{2} \left[1 - \sqrt{\frac{z_0}{z}} \right] / (z - z_0). \quad (14)$$

It may be noted that (12) and (13) are similar to expressions derived by Atkinson [24], but the author has not been able to demonstrate their equivalence.

Integral Representation of the Potentials

Thus far we have determined the field due to a single edge dislocation at z_0 in the presence of a crack. To apply this field to the branched-crack problem, we superpose a continuous distribution of dislocation singularities between 0 and 1 to simulate the branched crack, treating $\mu(\eta)d\eta$ as the infinitesimal Burger's vector, where now $\mu(\eta)$ is interpreted as $\partial/\partial\eta[u_r] + i[u_\theta]$. For simplicity, we first confine our attention to the singly branched crack in Fig. 1. The complex potentials are then given by

$$\left. \begin{aligned} \phi_2(z) &= \int_0^1 [\alpha(\eta) \ln(z - z_0) + \phi_R(z; z_0; \alpha(\eta))] d\eta, \\ \psi_2(z) &= \int_0^1 [\bar{\alpha}(\eta) \ln(z - z_0) - \alpha(\eta)\bar{z}_0/(z - z_0) + \psi_R(z; z_0; \alpha(\eta))] d\eta, \end{aligned} \right\} \quad (15)$$

where $z_0 = \eta e^{i\theta}$ and the density function $\alpha(\eta)$ remains as yet unknown.

Since the boundary conditions for the main crack are satisfied automatically, it remains only to insure that the stress boundary conditions for the branch be met. An integral equation will be derived from enforcing these conditions.

We note from (12), (13), and (15) that the asymptotic behavior of ϕ_2' for large $|z|$ is given by

$$\phi_2' = \mathcal{O}\left(\frac{1}{z^2}\right) \quad \text{as } |z| \rightarrow \infty. \quad (16)$$

This is different from the required asymptotic behavior of the solution ϕ' in (3). Consequently we set

$$\phi = \phi_2 + \phi_\infty, \quad \psi = \psi_2 + \psi_\infty, \quad (17)$$

where ϕ_∞, ψ_∞ are the potentials for the problem of a crack of length $2c$ in an infinite region under uniform tensile stress at infinity. The

solution of this problem is well known and is generated by the potentials

$$\left. \begin{aligned} \phi_\infty'(z) &= \frac{1}{2} \left[-\frac{1}{2} + \frac{z + c}{\sqrt{z(z + 2c)}} \right], \\ \psi_\infty'(z) &= \frac{1}{2} \left[1 + \frac{zc^2}{(\sqrt{z(z + 2c)})^3} \right]. \end{aligned} \right\} \quad (18)$$

The potentials ϕ, ψ in (17), once the boundary conditions for the branch have been satisfied, furnish the complete solution to the problem of the singly branched crack. The behavior at infinity is now taken care of by $\phi_\infty', \psi_\infty'$, which satisfy (3). Note that in the limit as $c/l \rightarrow \infty$, the potentials $\phi_\infty'(z), \psi_\infty'(z)$ in (18), upon nondimensionalization by $K_I^{(\infty)} = \sqrt{\pi c}$ according to (5), still combine (using 1)) to produce the correct far field behavior (4). This is not so in most of the previous work, where the corresponding potentials $\phi_\infty', \psi_\infty'$ are given by (3) which for large c/l still give uniform stress at infinity. Consequently difficulties arose in letting $c/l \rightarrow \infty$ since the problem of a uniformly stressed plate with a semi-infinite crack is improperly posed

while, equivalently, letting the branch-crack length go to zero ($l/c \rightarrow 0$) gives rise to numerical difficulties. This is related to the fact that the stress boundary conditions for the main crack are not satisfied automatically in most of the previous work.

Integral Equations

To satisfy the boundary conditions for the branch, we first represent the normal and shear stresses $\sigma_{\varphi\varphi}, \sigma_{\rho\varphi}$ on it in terms of ϕ' and ψ' . This yields

$$\sigma_{\varphi\varphi} + i\sigma_{\rho\varphi} = \phi' + \bar{\phi}' + e^{2i\theta}(\bar{z}\phi'' + \psi'), \quad (19)$$

where θ is the angle the branch makes with the crack-axis. The singular parts of ϕ' and ψ' will give different limits as the branched crack is approached from the left and the right. Using (13), (15), (17), and Plemelj's formulas, we obtain the stresses in (19) in terms of $\alpha(\eta)$ on the branched crack as

$$\sigma_{\varphi\varphi} + i\sigma_{\rho\varphi} = 2 \int_0^1 \frac{\bar{\alpha}(\eta)e^{i\theta}d\eta}{t - \eta} + \int_0^1 L_A(t, \eta; \alpha(\eta))d\eta + \sigma_{\varphi\varphi}^{(\infty)} + i\sigma_{\rho\varphi}^{(\infty)} = 0, \quad (20)$$

where L_A is the nonsingular part of the integral equation for the asymmetrically branched crack. The latter is given in terms of ϕ_R', ψ_R' , and $\alpha(\eta)$ in the Appendix. For the symmetrically and doubly symmetrical branched cracks, L_A is replaced by $L_S(t, \eta; \alpha(\eta))$ and $L_D(t, \eta; \alpha(\eta))$, respectively. The stresses $\sigma_{\varphi\varphi}^{(\infty)}, \sigma_{\rho\varphi}^{(\infty)}$ are associated with $\phi_\infty', \psi_\infty'$ and are known functions of z .

Symmetrically Branched Crack

The derivation of the integral equation for the singly branched crack may be readily modified to apply to the case of the symmetrically branched crack (Fig. 2). From the physical interpretation of $\alpha(\eta)$

in (8) it is clear how one should proceed to derive the integral equations for the remaining geometries considered in this study. For the symmetrically branched crack, the displacements in the lower half plane are, by symmetry, related to those in the upper half plane in the case of uniform tensile loading through

$$u_x(x, y) = u_x(x, -y), \quad u_y(x, y) = -u_y(x, -y), \quad (21)$$

i.e., the horizontal displacement is an even function of y and the vertical displacement u_y , an odd function of y . To obtain the singular solution for this problem, we have to first add another edge dislocation in the lower half plane at $\bar{z}_0$, consistent with the symmetry requirement of (21), so that the new potentials become

$$\phi_0 = f_1(z, z_0, \alpha) + f_1(z, \bar{z}_0, \bar{\alpha}), \quad \psi_0 = g_1(z, z_0, \alpha) + g_1(z, \bar{z}_0, \bar{\alpha}) \quad (22)$$

with f_1, g_1 defined in (8).

Recall that the term ϕ_R' , may be determined in such a way that the tractions resulting from ϕ_1, ψ_1 are zero on the main crack. To accomplish this, it is necessary to know the stresses $\sigma_{yy}^{(0)}(x, 0) - i\sigma_{xy}^{(0)}(x, 0)$ as in the singly branched crack case. These stresses can be inferred by inspection of (10) and (22). By adding (10) to its conjugate terms, we arrive at the following expression for the stresses on the main crack:

$$\begin{aligned} \sigma_{yy}^{(0)}(x, 0) - i\sigma_{xy}^{(0)}(x, 0) \\ = \frac{\alpha + \bar{\alpha}}{x - z_0} + \frac{\alpha + \bar{\alpha}}{x - \bar{z}_0} + \frac{\bar{\alpha}(z_0 - \bar{z}_0)}{(x - \bar{z}_0)^2} + \frac{\alpha(\bar{z}_0 - z_0)}{(x - z_0)^2} \Bigg\} \\ = -[p(x, z_0) + p(x, \bar{z}_0)] \end{aligned} \quad (23)$$

From (23), the term ϕ_R' can similarly be obtained by inspection

$$\phi_R' = -[(\alpha + \bar{\alpha})F(z, z_0) + (\alpha + \bar{\alpha})F(z, \bar{z}_0) + \bar{\alpha}(z_0 - \bar{z}_0)G(z, \bar{z}_0) + \alpha(\bar{z}_0 - z_0)G(z, z_0)], \quad (24)$$

where F, G are given as before in (13).

Taking into account the symmetry of the problem, we may restrict ourselves to the upper half plane. Note that the logarithmic term $\ln(z - \bar{z}_0)$ is regular in the upper half plane and the integral equation (20) remains valid but now the nonsingular part in terms of ϕ_R', ψ_R' is given in the Appendix.

Doubly Symmetrical Branched Crack

As before, we calculate the corresponding ϕ_R' for this problem with the singular potentials ϕ_0, ψ_0 given by

$$\begin{aligned} \phi_0 &= f_1(z, z_0, \alpha) + f_1(z, \bar{z}_0, \bar{\alpha}) \\ &\quad - f_1(z, -z_0 - 2c, \alpha) - f_1(z, -\bar{z}_0 - 2c, \bar{\alpha}), \\ \psi_0 &= g_1(z, z_0, \alpha) + g_1(z, \bar{z}_0, \bar{\alpha}) \\ &\quad - g_1(z, -z_0 - 2c, \alpha) - g_1(z, -\bar{z}_0 - 2c, \bar{\alpha}), \end{aligned} \quad (25)$$

where the functions f_1, g_1 are defined in (8). The corresponding ϕ_R' is

$$\begin{aligned} \phi_R' &= -[(\alpha + \bar{\alpha})F(z, z_0) + (\alpha + \bar{\alpha})F(z, \bar{z}_0) \\ &\quad + \alpha(\bar{z}_0 - z_0)G(z, z_0) + \bar{\alpha}(z_0 - \bar{z}_0)G(z, \bar{z}_0)] \\ &\quad + [(\alpha + \bar{\alpha})F(z, -z_0 - 2c) + (\alpha + \bar{\alpha})F(z, -\bar{z}_0 - 2c) \\ &\quad + \alpha(-\bar{z}_0 + z_0)G(z, -z_0 - 2c) \\ &\quad + \bar{\alpha}(-z_0 + \bar{z}_0)G(z, -\bar{z}_0 - 2c)]. \end{aligned} \quad (26)$$

Stresses at the Branched Crack Tip

We now write the unknown density function $\alpha(\eta)$ in (20) in the form

$$\alpha(\eta) = \frac{\beta(\eta)}{\sqrt{\eta(1-\eta)}} \quad (0 < \eta < 1). \quad (27)$$

This is a defining equation for $\beta(\eta)$. We anticipate that $\beta(\eta)$ is bounded at both ends $\eta = 0$ and $\eta = 1$ because near $\eta = 1$ the stresses behave like $\sigma_{ij} \sim \tilde{\sigma}_{ij}(\tilde{\rho})\tilde{\rho}^{-1/2}$ as $\tilde{\rho} \rightarrow 0$, where $\tilde{\rho}$ is the radial distance from the branched crack tip and the amplitudes $\tilde{\sigma}_{ij}(\tilde{\rho})$ are functions of the polar

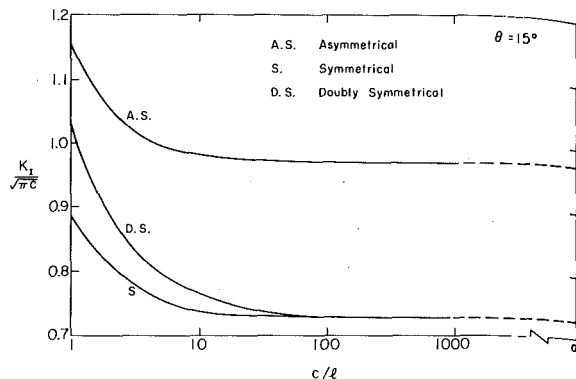


Fig. 4 Mode I stress-intensity factor versus c/l

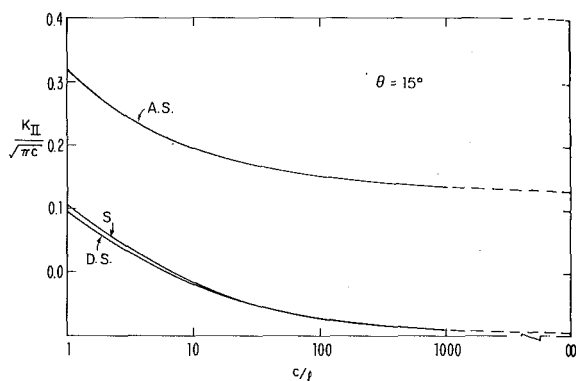


Fig. 5 Mode II stress-intensity factor versus c/l

angles $\tilde{\varphi}$ alone. Near $\eta = 0$, the stresses are like $\sigma_{ij} \sim \tilde{\sigma}_{ij}(\varphi)\rho^{-s}$ with $s \leq 1/2$.

The stresses near $t = 1$, but not on the crack, are given by

$$\sigma_{\varphi\varphi} + i\sigma_{\rho\varphi} \sim \int_0^1 \frac{\beta(\eta)e^{i\theta}d\eta}{\sqrt{\eta(1-\eta)}(t-\eta)} \sim 2\pi e^{i\theta} \frac{\beta(1)}{\sqrt{t-1}} \quad \text{as } t \rightarrow 1 \quad \text{at } \varphi = \theta. \quad (28)$$

The local stress-intensity factors at the branched crack tip are defined as

$$K_I + iK_{II} = \lim_{\rho \rightarrow 1^+} \sqrt{2\pi(\rho-1)} [\sigma_{\varphi\varphi}(\rho-1, \theta) + i\sigma_{\rho\varphi}(\rho-1, \theta)], \quad (29)$$

where (ρ, φ) are the polar coordinates with the point O as the origin. Using (28), we obtain

$$K_I + iK_{II} = (2\pi)^{3/2} e^{i\theta} \beta(1). \quad (30)$$

Thus once $\beta(1)$ is known from solving the integral equation, the stress-intensity factors can be evaluated on the basis of (30).

The integral equation (20) was solved numerically using a method developed by Erdogan, et al. [25], and the results are discussed in the next section.

Discussion and Concluding Remarks

Previous work on branched-crack problems has been limited mostly to finite values of c/l . In this paper, the range of c/l considered includes the limiting case $c/l \rightarrow \infty$. Figs. 4-9 show the results for the stress-intensity factors plotted against c/l for various angles of inclination θ . In the case of the singly branched crack, the present results agree with [8, 11, 12] for finite c/l and with [3, 4] in the limit as $c/l \rightarrow \infty$. For the symmetrically branched crack, the stress-intensity factors calculated for finite c/l in the present study were found to agree with those of [11, 12] but disagree with [10]. This does not support the

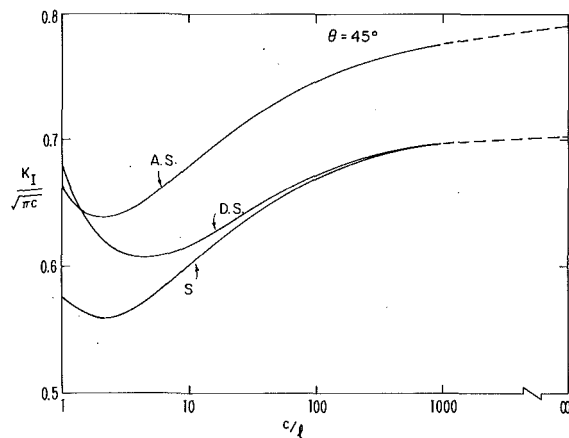


Fig. 6 Mode I stress-intensity factor versus c/l

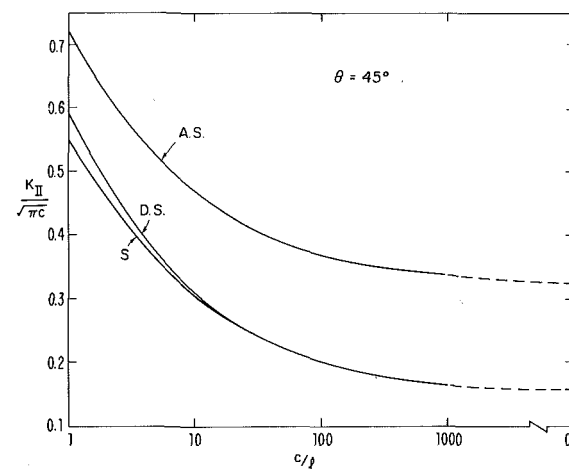


Fig. 7 Mode II stress-intensity factor versus c/l

suggestion in [10] that the results in [11, 12] may not be accurate. The results obtained in the present paper for the limiting case as $c/l \rightarrow \infty$ appear to be new.

In summary, the results for the stress-intensity factors in this study for the *singly branched crack* were found to be at variance with those of Hussain, et al. [7]. It should be added, however, that the calculations for the maximum energy release rate given in [7] agree with [3, 4, 13] and with this study, but disagree with Dudukalenko and Romalis [6] in the limiting case of $c/l \rightarrow \infty$. The present findings also confirm the work of Chatterjee [8], Kitagawa, Yuuki, and Ohira [11, 12], and Gupta [9] for finite c/l . For the *symmetrically branched crack*, the present study led to agreement with Theocaris and Ioakimidis [17], Kitagawa, et al. [11, 12] for finite c/l ; on the other hand, the present results differ from those of Andersson [1], Bilby, et al. [10], Cherepanov and Kuliev [21]. In the case of the *doubly symmetrical branched crack*, as far as one can gather from the graphical results given in [11, 12], the stress-intensity factors agree with those in [11, 12] to within 10 percent. Finally, we should mention that Vitek [26] recently computed the stress-intensity factors for the doubly symmetrical branched crack for finite c/l . Insofar as his calculations can be compared to the present results, the agreement is within 3 percent.

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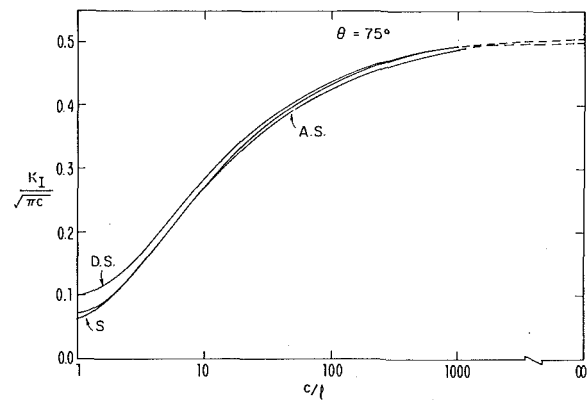


Fig. 8 Mode I stress-intensity factor versus c/l

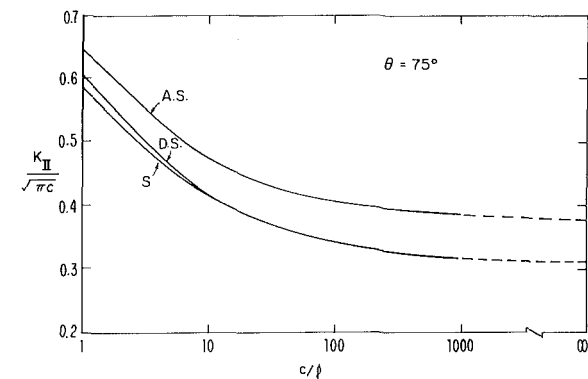


Fig. 9 Mode II stress-intensity factor versus c/l

References

- Andersson, H., "Stress-Intensity Factors at the Tips of a Star-Shaped Contour in an Infinite Tensile Sheet," *Journal of the Mechanics and Physics of Solids*, Vol. 17, 1969, pp. 405-417.
- Andersson, H., Erratum on (1), *Journal of the Mechanics and Physics of Solids*, Vol. 18, 1970, p. 437.
- Palaniswamy, K., and Knauss, W. G., "On the Problem of Crack Extension in Brittle Solids Under General Loading," Report SM 74-8, Graduate Aeronautical Lab., Caltech.
- Bilby, B. A., and Cardew, G. E., "The Crack With a Kinked Tip," *International Journal of Fracture*, 1975, pp. 708-711.
- Khrapkov, A. A., "The First Basic Problem for a Notch at the Apex of an Infinite Wedge," translated by I. Sneddon, *International Journal of Fracture Mechanics*, Vol. 7, No. 4, 1971, pp. 373-382.
- Dudukalenko, V. V., and Romalis, N. B., "Direction of Crack Growth Under Plane-Stress-State Conditions," *Izv. AN SSSR, Mekhanika Tverdogo Tela*, Vol. 8, No. 2, 1973, pp. 129-136.
- Hussain, M. A., Pu, S. L., and Underwood, G., "Strain Energy Release Rate for a Crack Under Combined Mode I and Mode II," Technical Report, Benet Weapons Lab., Aug. 1973.
- Chatterjee, S. N., "The Stress Field in the Neighborhood of a Branched Crack in an Infinite Sheet," *International Journal of Solids and Structures*, Vol. II, 1975, pp. 521-538.
- Gupta, G. D., "Strain Energy Release Rate for Mixed Mode Crack Problem," ASME Paper No. 76-WA/PVP-7.
- Bilby, B. A., Cardew, G. E., and Howard, I. C., "Stress Intensity Factors at the Tips of Kinked and Forked Cracks," *Fracture 1977*, editor, Taplin, D. M. R., Vol. 3, University of Waterloo Press, Canada, 1977, pp. 197-200.
- Kitagawa, H., and Yuuki, R., "Analysis of Branched Cracks Under Biaxial Stresses," *Fracture 1977*, editor, Taplin, D. M. R., Vol. 3, University of Waterloo, Canada, 1977, pp. 201-211.
- Kitagawa, H., and Yuuki, R., and Ohira, T., "Crack-Morphological Aspects in Fracture Mechanics," *Eng. Fract. Mech.*, Vol. 7, 1975, pp. 515-529.
- Wu, C. H., "Fracture Under Combined Loads by Maximum-Energy-Release-Rate Criterion," presented at the Joint Applied Mechanics Fluid Engineering and Bioengineering Conference, 1977.
- Erdogan, F., and Sih, G. E., "On the Crack Extension in Plates Under Plane Loading and Transverse Shear," *ASME Journal of Basic Engineering*, Vol. 85, Dec. 1963, pp. 519-527.

- 15 Sih, G., "Some Basic Problems in Fracture Mechanics and New Concepts," *Eng. Fract. Mech.*, Vol. 5, 1973, pp. 365-377.
- 16 Bui, H. D., "Approximate Path Independent Integrals in the Plane Problem of Cracks and Associated Anti-plane Shear Problems," *Archives of Mech.*, Vol. 28, 1976, pp. 649-659.
- 17 Theocaris, P. S., and Ioakimidis, N., "The Symmetrically Branched Crack in an Infinite Elastic Medium," *Journal of Applied Mathematics and Physics (ZAMP)*, Vol. 27, 1976, pp. 801-814.
- 18 Datsyshin, A. P., and Savruk, M. P., "A System of Arbitrarily Oriented Cracks in Elastic Solids," *Journal of Applied Mathematics and Mechanics (PMM)*, 38, pp. 677-686 (1974).
- 19 Greif, R., and Sanders, J. L., "The Effect of a Stringer on the Stress in a Cracked Sheet," *ASME JOURNAL OF APPLIED MECHANICS*, Vol. 32, 1965, pp. 59-66.
- 20 Rice, J. R., "Mathematical Analysis in the Mechanics of Fracture," Chapter 3 in *Fracture*, vol. II, edited by Liebowitz, H., Academic Press, 1968, pp. 192-311.
- 21 Cherepanov, G. P., and Kuliev, V. D., "On Crack Twinning," *International Journal of Fracture*, Vol. 11, 1975; pp. 29-38.
- 22 Muskhelishvili, N. L., *Some Basic Problems of Mathematical Theory of Elasticity*, P. Noordhoff, 1958.
- 23 Erdogan, F., "On the Stress Distributions in Plates With Collinear Cuts Under Arbitrary Loads," *Proceedings of the Fourth U. S. National Congress of Applied Mechanics*, Vol. I, 1962, pp. 547-554.
- 24 Atkinson, C., "The Interaction Between a Dislocation and a Crack," *International Journal of Mechanics*, Vol. 2, 1966, pp. 567-574.
- 25 Erdogan, F., Gupta, G. D., and Cook, T. S., "Numerical Solution of Singular Integral Equations," Chapter 7 in *Mechanics of Fracture I*, edited by Sih, G. C., 1973.
- 26 Vitek, V., "Plane Strain Stress-Intensity Factors For Branched Cracks," *International Journal of Fracture*, Vol. 13, No. 4, Aug. 1977, pp. 481-501.

APPENDIX

We have the nonsingular expression in equation (20) in terms of ϕ_R' , ψ_R' (see (13)) as follows:

Asymmetrically Branched Crack

$$L_A(t, \eta; \alpha) = \phi_R'(z) + \bar{\phi}_R'(\bar{z}) + e^{2i\theta}[(\bar{z} - z)\phi_R''(z) - \phi_R'(z) + \bar{\phi}_R'(z)], \quad (31)$$

where (see (12)) $z = te^{i\theta}$ and $z_0 = \eta e^{i\theta}$.

Symmetrically Branched Crack

$$L_s(t, \eta; \alpha(\eta)) = \phi_R'(z) + \bar{\phi}_R'(\bar{z}) + e^{2i\theta}(\bar{z} - z)\phi_R''(z) + 2\alpha(\eta)e^{i\theta} \operatorname{Re} \left(\frac{e^{i\theta}}{z - \bar{z}_0} \right) + \overline{\alpha(\eta)} \frac{z_0 2i \sin 2\theta}{(z - \bar{z}_0)^2}, \quad (32)$$

where now $\phi_R'(z)$ is given by (24).

Doubly Symmetrical Branched Crack

$$L_D(t, \eta; \alpha(\eta)) = \phi_R'(z) + \bar{\phi}_R'(\bar{z}) + e^{2i\theta}(\bar{z} - z)\phi_R''(z) + 2\alpha(\eta)e^{i\theta} \operatorname{Re} \left(\frac{e^{i\theta}}{z - \bar{z}_0} \right) + \overline{\alpha(\eta)} \frac{z_0 2i \sin 2\theta}{(z - \bar{z}_0)^2} - \left[2\overline{\alpha(\eta)}e^{i\theta} \operatorname{Re} \left(\frac{e^{i\theta}}{z + z_0'} \right) + \alpha(\eta)2c \frac{(1 - e^{2i\theta})}{(z + z_0')^2} \right] - \left[2\alpha(\eta)e^{i\theta} \operatorname{Re} \left(\frac{e^{i\theta}}{z + \bar{z}_0'} \right) + \overline{\alpha(\eta)} \frac{2c(1 - e^{2i\theta})}{(z + \bar{z}_0')^2} - \frac{2\overline{\alpha(\eta)} iz_0 \sin 2\theta}{(z + \bar{z}_0')^2} \right], \quad (33)$$

where ϕ_R' is given by (26) and $z_0' = z_0 + 2c$.